

The Accuracy Illusion: How Overlapping Samples Manufacture Confidence, Not Skill, in Intraday Index Forecasting

Rachit Ranka

Department of Computer Science Engineering
Chandigarh University
Mohali, India
rankarachit5@gmail.com

Aaditya Saini

Department of Computer Science Engineering
Chandigarh University
Mohali, India
aadityasaini899@gmail.com

Abstract—Intraday forecasting systems routinely report directional hit-rates between 60% and 85%, and those figures move real capital. We show they are not valid estimates of skill. A k -bar-ahead forecast issued every bar shares $k-1$ of its k outcome bars with its neighbour; when the model’s calls also persist, as they do on a live system we instrument, the reported hit-rate acquires a variance far larger than its nominal sample size implies. The reading is not biased upward. Its stated precision is. A forecaster with *exactly zero skill*, simulated against real NSE index bars, posts a $\geq 71\%$ session once in 48 while reporting an interval $1.68\times$ too narrow, rising to $3.40\times$ at the per-minute density these systems publish.

We give the closed form behind this, the classical design effect with clusters given by runs of identical calls, and a calibrated interval built on it that covers at 93% where the conventional one covers at 73%. Applied to our own results it leaves nothing: across 48 model-index-horizon tests, 16 clear $p < 0.05$ scored per bar and zero clear it per event or at an effective sample size. A premium-selling strategy we had published as this project’s one measurable edge fails the same correction. Break-even accuracy derived from measured move distributions is 71.5% at thirty minutes against a best achieved 51.1% on the same index, and at five minutes no accuracy suffices under a symmetric bracket. Code and raw results are public.

Index Terms—overlapping observations, backtest overfitting, directional forecasting, negative results, reproducibility, index derivatives

I. INTRODUCTION

A system reporting 71% directional accuracy is, on its face, extraordinary. Work that controls for scoring unit, costs and multiple testing places realistic out-of-sample intraday index accuracy close to a coinflip [1], [2], while surveys of the applied literature catalogue many higher figures reported without those controls [3].

This paper began as an attempt to build such a system. Over fourteen months we ran a live multi-horizon forecasting platform for Indian index derivatives, and it reported 71%. That number was wrong. The specific session producing it cannot be re-audited, because the portion of the

log holding it was lost before this analysis; every figure below comes from sessions that survive.

The econometrics of overlapping observations is not new [4], [5]. Boudoukh et al. [6] show that overlap manufactures apparent long-horizon predictability which vanishes under correct inference. What that literature does not document is the magnitude of the distortion in a now-ubiquitous practice: the every-minute multi-horizon dashboard, where overlap meets a second ingredient, prediction persistence, and is then amplified across a panel.

Contributions. (i) We measure the artefact as it occurs, showing that overlap alone changes nothing and that inflation requires overlap *and* persistent predictions, with persistence calibrated from a production log. (ii) We derive the effective sample size of such a stream, show it is governed by $\mathbb{E}[L^2]/\mathbb{E}[L]$ over run lengths, and calibrate an interval that covers at its nominal rate. (iii) We apply it to our own results and to a complete control set — shuffled labels, injected positive controls, eight architectures, a block-bootstrap null — and report that nothing survives. (iv) We invert accuracy into a required-versus-achieved framing that shows the gap is not close. Code, raw results and the free-data corpus are at <https://github.com/rachitranka25/accuracy-illusion>.

II. RELATED WORK

Hansen and Hodrick [4] and Richardson and Smith [5] characterise inference under overlapping returns; Newey and West [7] give the standard correction. That work concerns regression coefficients. We apply the same logic to what practitioners publish, the session hit-rate, and the correction we reach is the classical design effect [8], [9] with clusters given by runs of identical calls. The contribution is its application to forecast streams, not the statistic.

On multiplicity, Harvey et al. [10] argue the conventional $t > 2$ threshold is indefensible after decades of factor mining. Bailey and López de Prado [11] introduce the Deflated Sharpe Ratio and Bailey et al. [12] the Probability of Backtest Overfitting; White’s Reality Check [13], its refinement by Hansen [14] and Benjamini–Hochberg [15]

TABLE I
FREE-DATA CORPUS. ALL COLLECTORS ARE PUBLIC.

Asset	Period	Size
BANKNIFTY 5-min bars	2025-02 – 2026-07	26 265
NIFTY 50 5-min bars	2025-02 – 2026-07	26 267
3 further indices, 5-min	2025 – 2026	9.6k–14k
Option chain EOD + OI	2024-07 – 2026-07	1 382 738
Long-history option chain	2016-05 – 2026-05	2 181 d
Participant-wise OI	2024-07 – 2026-07	515 d
Index futures (real volume)	2024-01 – 2026-07	639 d

provide complementary controls. Sullivan et al. [2] apply the Reality Check to a universe of trading rules, the direct precedent for our strategy sweep.

On machine learning for finance, Fischer and Krauss [16] and Krauss et al. [17] report headline performance gross of costs; Gu et al. [18] give a comprehensive comparison with modest out-of-sample R^2 . Zhang et al. [19] and Kolm et al. [20] demonstrate genuine short-horizon alpha, both from message-level exchange data rather than bars — the distinction that matters here. Across that set, none reports a multiple-testing correction or a shuffled-label null control in the form used here, and only one releases code and results in part, and the two S&P 500 studies quote headline performance gross of transaction costs. The scoring unit also differs: daily and monthly studies aggregate over non-overlapping periods, where the artefact documented here does not arise, while the limit-order-book studies score per event. Per-bar scoring of a multi-bar horizon, the condition under which it does arise, is characteristic of intraday dashboards rather than of the published work we cite.

III. DATA AND METHOD

All data is obtainable free from a retail brokerage account and public exchange archives (Table I). Three practical observations. Index spot prints zero volume, so a feature computed as a “VWAP” on spot is an unweighted expanding mean; real volume exists only on futures. The exchange’s *live* option-chain API is geo-restricted while its static end-of-day archive is not, which turned an assumed constraint into a decade of data. And brokers authenticating by locally computed TOTP permit collection from any jurisdiction, where SMS-based flows do not.

Every accuracy passes through one harness. Splits are expanding-window walk-forward with the k bars whose labels overlap the test window purged from training [21]. Accuracies carry Clopper–Pearson [22] intervals, exact binomial p -values and the Pesaran–Timmermann statistic [23], which is the appropriate null for directional accuracy. Every model is additionally trained on randomly permuted labels; if real and shuffled scores coincide, the pipeline is reading noise. A synthetic feature of known strength is injected as a positive control, since a pipeline that cannot recover planted signal cannot be trusted to report a null. Strategy returns use non-overlapping k -bar holds with explicit costs.

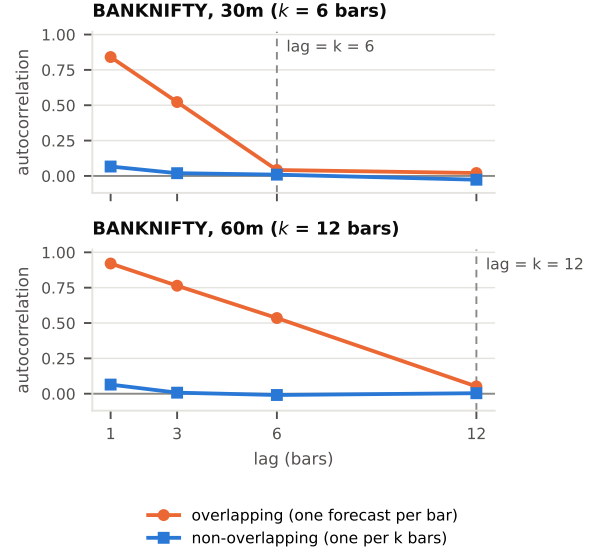


Fig. 1. Forward-return autocorrelation, overlapping versus non-overlapping, at two horizons. The overlapping series dies at exactly lag k in both panels.

Units. All analysis is on 5-minute bars, so the 30-minute horizon is $k = 6$ and a session yields $n = 69$ forecasts. The production system emits a forecast every *minute*; pooled, its run lengths average 15.2 minutes with a longest of 263; the pooled mean run in 5-minute bars is 3.4.

IV. THE MECHANISM

A. *Overlap is arithmetic*

Figure 1 shows the autocorrelation of the k -bar forward return two ways. On overlapping windows it reaches 0.841 at lag 1 and decays to approximately zero at *exactly* lag k ; sampled so no two observations share a bar it is 0.067 and indistinguishable from zero thereafter. The pattern reproduces at $k = 6$ and $k = 12$, with the decay landing on lag 6 and lag 12. The correlation is a property of the sampling scheme, not the market.

B. *Persistence is the second ingredient*

Overlap alone is insufficient. If predictions are drawn independently each bar, the hit indicator is i.i.d. Bernoulli regardless of how autocorrelated the outcome is. Our first simulation made exactly this error and correctly found nothing. Real models do not re-decide every bar: measured on the production log at the 30-minute horizon, BANKNIFTY calls persist for a mean of 25.9 minutes and NIFTY 50 for 10.3, against a mean run of one for a model that re-drew each bar. When both the call and the outcome persist, one trending stretch is recorded as dozens of hits.

TABLE II

A FORECASTER WITH EXACTLY ZERO SKILL, 30-MINUTE HORIZON, SCORED PER BAR VERSUS PER EVENT. BANKNIFTY; NIFTY 50 WITHIN 0.6 POINTS ON EVERY ROW.

	Per bar	Per k bars
Reported sample size per session	69	12
Stated binomial 95% CI	± 11.8 pts	± 28.3 pts
Realised spread across sessions	± 19.9 pts	± 28.0 pts
Understatement factor	$1.68\times$	$0.99\times$
$P(\text{session} \geq 60\%)$	16.2%	18.9%
$P(\text{session} \geq 71\%)$	2.1%	7.1%

C. What a zero-skill forecaster reports

We simulate a fair coin held for a run length drawn from that distribution, evaluated against real bars and scored as a dashboard scores it (Table II).

The table states a point easily read backwards. Per-bar scoring does *not* produce extreme readings more often than honest scoring; it produces them *less* often. What it does is report them with an interval $1.68\times$ too narrow, because the quoted n of 69 is a fiction. Overlap manufactures unwarranted *confidence*, not high hit-rates, and it is that false precision which misleads.

D. At the real granularity, and across a panel

The production system forecasts every minute, so a session yields about 313 scored forecasts rather than 69. Taking the outcome series from the log itself and simulating the same zero-skill forecaster, the understatement factor rises to $3.40\times$ while extreme sessions become slightly *rarer*. Every 5-minute-bar figure here therefore understates the artefact.

A dashboard also watches more than one number: two indices at four horizons is eight hit-rates per session. Simulating the panel jointly on the same sessions gives $P(\exists \text{series} \geq 71\%) = 17.1\%$, against 17.2% under an independence assumption — adequate here because a zero-skill forecaster’s hits are near-independent across series even when outcomes are not. Either way, about one session in six.

V. A CALIBRATED ESTIMATOR

A. Effective sample size

Let $h_t = \mathbf{1}\{\hat{y}_t = y_t\}$ and $\hat{p} = n^{-1} \sum_t h_t$. Practitioners quote $\text{Var}(\hat{p}) = p(1-p)/n$, valid only for independent h_t . In general

$$\text{Var}(\hat{p}) = \frac{p(1-p)}{n} \left[1 + 2 \sum_{j=1}^{n-1} \left(1 - \frac{j}{n}\right) \rho_j \right], \quad (1)$$

with $\rho_j = \text{Corr}(h_t, h_{t+j})$. Because the dependence here is block-shaped, this collapses. If the sequence decomposes into runs of length L_i within which h_t is constant, then

$$n_{\text{eff}} = \frac{n}{\sum_i L_i^2/n} \xrightarrow{n \rightarrow \infty} \frac{n \mathbb{E}[L]}{\mathbb{E}[L^2]}. \quad (2)$$

TABLE III

COVERAGE OF A NOMINAL 95% INTERVAL ON A SESSION HIT-RATE. BANKNIFTY, 30-MINUTE HORIZON, NOMINAL $n = 69$; NIFTY 50 AGREES WITHIN 2 POINTS. HAC-L GIVES THE BARTLETT KERNEL A RUN-STRUCTURE BANDWIDTH; DERIVED DIVIDES THE DESIGN EFFECT BY THE ALTERNATION FACTOR OF (3).

Skill	Naive	HAC	HAC-L	Design	Deriv.	n_{eff}
50%	73.4	86.2	88.0	99.7	93.2	11.9
54%	73.9	86.6	88.1	99.6	93.5	11.7
59%	73.0	85.2	86.8	99.6	93.5	10.9

This is the classical design effect [8], [9]; what is new is the observation that a forecast stream has exactly this structure and that nobody reports it. The inflation is governed by $\mathbb{E}[L^2]$, so the tail matters, not the mean.

B. Validation by coverage

Table III reports coverage against a forecaster of known skill whose calls persist per the measured distribution, with true skill measured from the generative process rather than assumed.

HAC and a block bootstrap (86.2%, omitted for space) both under-cover by about nine points. The obvious diagnosis is bandwidth: the automatic Bartlett rule gives 3.7 bars at $n = 69$, close to the pooled mean run of 3.4, while the tail reaches 53. We tested that rather than asserting it — a run-structure bandwidth recovers one to two points and no more, so bandwidth was part of the problem and not the whole of it, and we report the repair though it mostly fails. The design effect (2) removes the failure mode entirely and over-corrects in the other direction, reaching 99.7%. That over-statement is derivable rather than incidental. Treating runs as independent blocks ignores that a binary sequence’s runs alternate deterministically: a run of hits is always followed by a run of misses. For geometric run lengths of mean μ ,

$$\frac{\mathbb{E}[L^2]}{\mathbb{E}[L]} = 2\mu - 1, \quad \text{exact } \frac{1 + \rho}{1 - \rho} = \mu - 1, \quad (3)$$

so (2) over-states the dependence by $(2\mu - 1)/(\mu - 1)$, a quantity that tends to 2 for long runs. Evaluated per session it averages 2.44 at a mean hit-run length of $\mu = 3.53$; the residual is the departure of real run lengths from geometry.

Dividing by that factor, computed from each sample’s own mean run length, introduces no free constant and gives 93.1–93.5% coverage across both indices and all three skill levels. We report it as the recommended estimator, and the raw design effect as the conservative bound.

C. Reliability, and a real session

Equation (2) depends on a tail-driven second moment estimated from a few dozen runs. Measured against a pooled reference of 10.6, a single session’s n_{eff} lands between $0.63\times$ and $1.73\times$ nine times in ten. So n_{eff} should be estimated from a run of sessions, and treated as an order of magnitude from one.

TABLE IV

MODEL-INDEPENDENT CEILING TEST, 30-MINUTE HORIZON, FIVE INDICES. TRAINING ACCURACY IS 100.0% ON EVERY ROW. POSITIVE CONTROLS HAVE RECOVERABLE ACCURACIES OF ABOUT 55% AND 65% BY CONSTRUCTION.

Index	n	Real	Shuf.	Maj.	$s=.10$	$s=.30$
BANKNIFTY	16 856	50.5	50.2	49.3	52.4	63.5
NIFTY 50	16 853	50.0	50.2	49.8	52.0	63.7
FINNIFTY	8 988	50.2	50.5	49.4	52.4	62.5
MIDCPNIFTY	8 309	50.2	50.8	49.7	53.3	62.0
SENSEX	6 144	49.7	50.1	49.0	52.5	62.1

That matters immediately. Applying the estimators to the production log, a BANKNIFTY session reporting 57.1% has a naive interval excluding 50% and a corrected interval of [34.4, 79.9]. The best NIFTY 50 session, 69.6% over 335 forecasts, has a corrected interval of [50.0, 89.1] — which excludes 50%, but at the 5th percentile of the n_{eff} uncertainty becomes [45.1, 94.0] and does not. The lower bound is smaller than the estimator’s own noise, so the session supports no claim. An earlier draft reported that it did.

Per-event scoring is unbiased and needs no estimator, and is what we use for every backtest here. It is insufficient for a live stream for three reasons: a dashboard cannot wait k bars or discard forecasts already shown; subsampling throws away 83% of the observations; and which of the k offsets is chosen moves the same session by 7 to 20 points, so an analyst free to pick the offset is free to pick the answer.

VI. PREDICTABILITY UNDER CORRECT SCORING

A. Is there anything to predict?

An unconstrained forest was fitted until it memorised the training data, on five indices across two exchanges (Table IV). Each memorises completely and generalises at chance; permuting labels moves accuracy by 0.3 to 0.7 points; no real-label result is distinguishable from a coinflip ($p = 0.22$ to 0.91). Both positive controls are recovered everywhere at $p < 0.001$, so the pipeline detects signal whenever signal is present.

B. Eight model families, three horizons

Eight families were trained on identical splits, features, seeds and controls: logistic regression, gradient boosting, LightGBM [24], XGBoost [25], k -nearest neighbours, an LSTM [26], a Transformer [27] and a compact Temporal Fusion Transformer [28]. Table V scores every one three ways.

Scored per bar, XGBoost reaches 51.3% on BANKNIFTY at $p = 0.001$. That was, in an earlier draft, our one surviving positive result, described as Bonferroni-robust. It does not survive our own correction: re-scored one observation per 6 bars it is 50.4% at $p = 0.692$, and widening intervals to n_{eff} puts 50% inside every one. Across all 48 tests the count falls from 16 to zero. The count of apparent discoveries rises with horizon,

TABLE V

MODEL FAMILIES SIGNIFICANT AT $p < 0.05$, BY HORIZON AND SCORING RULE. EIGHT FAMILIES PER CELL, EXCLUDING THE MAJORITY-CLASS BASELINE.

Horizon, index	Per bar	At n_{eff}	Per event
15 m BANKNIFTY	0	0	0
15 m NIFTY 50	2	0	0
30 m BANKNIFTY	3	0	0
30 m NIFTY 50	1	0	0
60 m BANKNIFTY	5	0	0
60 m NIFTY 50	5	0	0
Total (48 tests)	16	0	0

from none at 15 minutes on BANKNIFTY to five at 60 on both indices, exactly as it should if the artefact scales with k ; the count of surviving discoveries stays at zero.

Table V also contains its own diagnostic. The NIFTY 50 Transformer sits at 48.4% with $p < 0.001$, which at the nominal n is 4.3 standard errors *below* chance where the largest $|z|$ across 48 tests should land near 2.9–3.2. That is not a broken model; it is the variance inflation surfacing inside our own results, and at n_{eff} the anomaly vanishes.

C. Eighty-six strategies

Separately, 86 parameterised rules — momentum, mean reversion, moving-average crosses, RSI, Bollinger, break-out, gap, volatility regime, time-of-day and streak variants — were evaluated on non-overlapping 30-minute holds at 4 bp cost with a one-sided test for profit. Two have a positive Sharpe; none is significantly profitable, before or after Benjamini–Hochberg. The Deflated Sharpe Ratio returns an implausible null benchmark of 7.77 here, because trial Sharpe dispersion is dominated by deterministic turnover differences rather than noise, so we replaced it with a circular block bootstrap [29] that resamples forward returns while holding each strategy’s positions fixed. The best observed Sharpe is 0.21 against a null best-of-86 median of 0.44, with $P(\text{null} \geq \text{observed}) = 0.658$: the winner is worse than what selection alone produces. PBO via CSCV [12] is degenerate here and reported as such.

VII. ECONOMIC SIGNIFICANCE

For a symmetric bracket with target and stop at the typical move m and round-trip cost c , expectancy is $m(2p - 1) - c$, so break-even requires

$$p^* = \frac{1}{2} + \frac{c}{2m}. \quad (4)$$

The ratio c/m is the entire problem (Table VI).

Two consequences (Figure 2). At the shortest horizon the median move is smaller than the round trip, so under a symmetric bracket p^* exceeds 100% and no accuracy suffices. That conclusion is specific to the bracket: moves are strongly right-skewed, so a hold-to-horizon rule requires 86% on BANKNIFTY and 93% on NIFTY 50 instead — not impossible, merely far beyond anything here or in the literature. We report both, and the point survives either way.

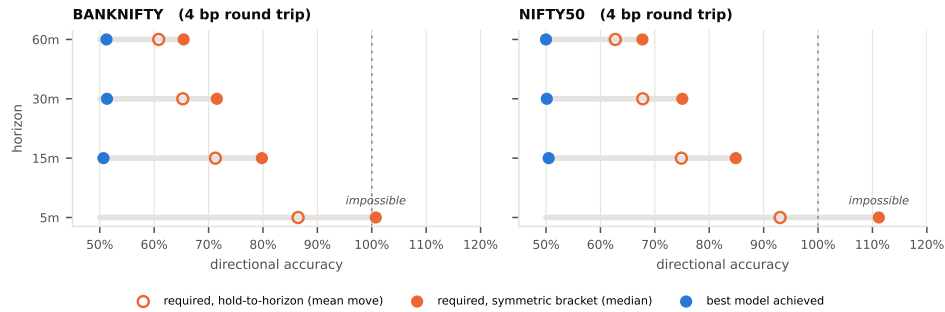


Fig. 2. Break-even accuracy against the best per-event accuracy any model family reached, by horizon and index, under both payoff models. Points beyond the dashed line at 100% are horizons at which no directional accuracy is sufficient under a symmetric bracket.

TABLE VI

BREAK-EVEN ACCURACY AT 4 BP ROUND-TRIP COST. p_{MED}^* ASSUMES A SYMMETRIC BRACKET AT THE MEDIAN MOVE; p_{MEAN}^* IS THE REQUIREMENT FOR A HOLD-TO-HORIZON RULE. ACHIEVED IS THE BEST PER-EVENT ACCURACY OF THE EIGHT FAMILIES.

Horizon	Med.	Mean	p_{med}^*	p_{mean}^*	Ach.
<i>BANKNIFTY</i>					
5 min	22.1	30.8	>100	86.4	—
15 min	37.7	52.9	79.8	71.3	51.3
30 min	52.3	73.8	71.5	65.3	51.1
60 min	72.9	103.8	65.4	60.8	52.2
<i>NIFTY 50</i>					
5 min	8.1	11.5	>100	93.0	—
15 min	14.1	19.8	84.9	74.9	50.5
30 min	19.7	27.8	75.1	67.8	51.2
60 min	27.7	38.5	67.7	62.7	50.1

Second, instrument choice can make the requirement unreachable. At 4 bp BANKNIFTY needs 71.5% at thirty minutes; at 8 bp, 93.0%; at the 20 bp a retail ATM option round trip costs over that horizon, $c > m$ and $p^* > 1$. The instrument most retail participants use, over the horizon they hold, has no directional accuracy that makes it profitable.

This also reframes the claims the paper is about. A genuine 85% would clear break-even by 13.5 points on BANKNIFTY futures and remain unprofitable on options. Such a claim is therefore either an artefact of Section IV or true and far less valuable than it sounds.

VIII. WHAT IS PREDICTABLE INSTEAD

These results went through the same harness, and what it leaves is narrow enough to state precisely: *none of this project's findings survives as a finding of ours.*

Volatility. A binary magnitude target reaches 70.8% per bar and 71.3% per event on BANKNIFTY (72.0 and 72.1 on NIFTY 50), with shuffled controls at 52–54%. That it is untouched by the correction which destroyed the directional result matters, since an instrument flattening everything would be useless. But a persistence rule already scores 67.4%, the model's edge is +3.3 to +3.7 points per event, and at n_{eff} the model and baseline intervals overlap. This is a confirmation of volatility clustering [30], [31], not a discovery of ours.

Variance risk premium, withdrawn. Short ATM straddles held to expiry, on real end-of-day premiums, across four entry points: no configuration is significantly profitable on either index, before or after Benjamini–Hochberg, and no bootstrap interval on the mean excludes zero on the profitable side. Recentring each configuration and resampling gives a best-of-four null median of 17.8 against an observed 12.1 on NIFTY ($P = 0.68$). NSE also moved BANKNIFTY options from weekly to monthly expiry mid-sample, so the two indices were never comparable: split on expiry spacing, the weekly sub-sample returns -67.5 points per trade and the monthly $+57.5$. The premium is real and documented [32], [33]; our measurement of it as an edge is withdrawn.

Option-chain positioning. Over 2181 days, after regressing out the price component, six of eight residual features clear $p < 0.05$ uncorrected and *one* survives Benjamini–Hochberg: the daily change in put–call open-interest ratio ($\text{IC} = 0.059$, $t \approx 2.8$, $\approx 51\%$ hit-rate), which is also the only feature strengthening in the second half. Flow beats position. Nothing in the price-only set clears $p < 0.05$ at all.

IX. LIMITATIONS

The alternation correction assumes geometric run lengths; real runs depart from that, which is why coverage lands near 93 % rather than 95 %. Its input is imprecisely measured from a single session. Two indices carry the primary intraday analysis over seventeen months in one market; the ceiling test replicates across five but the model zoo and strategy sweep do not. Break-even is computed for two payoff models, neither of which models the path, so the probability of touching a stop before a target is unaccounted. Some figures in our research record come from scripts not retained; two that were re-run did not reproduce and are withdrawn in place. Index spot carries no volume, so VWAP-derived features are misnamed. Our Temporal Fusion Transformer is a compact variant and should not be read as the published architecture [28]. Finally, the order-flow hypothesis, where the literature locates genuine short-horizon alpha [19], [20], is untested here for want of history.

X. CONCLUSION

The headline metric of intraday directional forecasting is not merely noisy. Its stated precision is systematically wrong, by enough to make a coinflip look like a discovery: a zero-skill forecaster posts a $\geq 71\%$ session once in 48, and once every six sessions somewhere on a modest panel, while reporting an interval $1.68\times$ too narrow at 5-minute granularity and $3.40\times$ too narrow at the density these systems publish.

Applied to our own work the correction leaves nothing standing. Sixteen of 48 model-index-horizon tests clear $p < 0.05$ per bar and none clears it per event. Eighty-six strategies produce a winner worse than a bootstrap null. Premium selling, which we had published as this project's one measurable edge, fails the same correction. The one untouched result confirms volatility clustering rather than finding anything.

So the honest summary is that this project found nothing tradeable, and its value lies in how thoroughly that was established. What we offer is a method: an effective sample size for overlapping persistent forecasts, a calibrated interval built on it, and a framing that turns an accuracy figure into a question about cost. Reporting practice should change to match, and the change is cheap. Scoring unit, effective sample size, multiplicity correction, a shuffled-label control and an explicit cost model are not optional refinements to an accuracy figure. Without them the figure does not mean anything, and we know that because we published several that did not.

REPRODUCIBILITY AND DATA AVAILABILITY

Every number here is regenerated by a script in the public repository, with the raw output of each committed alongside it, at <https://github.com/rachitranga25/accuracy-illusion>. That repository holds the project's complete research record, including the positive results withdrawn above. Experiments use a fixed seed. Collectors for every source in Table I are included, so the free-data corpus can be rebuilt.

SCOPE AND DISCLAIMER

This is a measurement study, not investment advice. Its central finding is that the reported accuracy of intraday directional systems is not a valid estimate of skill, a conclusion we apply first to our own published claims.

REFERENCES

- [1] A. Timmermann and C. W. J. Granger, "Efficient market hypothesis and forecasting," *Int. J. Forecast.*, vol. 20, no. 1, pp. 15–27, 2004.
- [2] R. Sullivan, A. Timmermann, and H. White, "Data-snooping, technical trading rule performance, and the bootstrap," *J. Finance*, vol. 54, no. 5, pp. 1647–1691, 1999.
- [3] O. B. Sezer, M. U. Gudelek, and A. M. Ozbayoglu, "Financial time series forecasting with deep learning: A systematic literature review 2005–2019," *Appl. Soft Comput.*, vol. 90, art. 106181, 2020.
- [4] L. P. Hansen and R. J. Hodrick, "Forward exchange rates as optimal predictors of future spot rates: An econometric analysis," *J. Polit. Econ.*, vol. 88, no. 5, pp. 829–853, 1980.
- [5] M. Richardson and T. Smith, "Tests of financial models in the presence of overlapping observations," *Rev. Financ. Stud.*, vol. 4, no. 2, pp. 227–254, 1991.
- [6] J. Boudoukh, M. Richardson, and R. F. Whitelaw, "The myth of long-horizon predictability," *Rev. Financ. Stud.*, vol. 21, no. 4, pp. 1577–1605, 2008.
- [7] W. K. Newey and K. D. West, "A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix," *Econometrica*, vol. 55, no. 3, pp. 703–708, 1987.
- [8] L. Kish, *Survey Sampling*. New York, NY, USA: Wiley, 1965.
- [9] B. R. Moulton, "An illustration of a pitfall in estimating the effects of aggregate variables on micro units," *Rev. Econ. Stat.*, vol. 72, no. 2, pp. 334–338, 1990.
- [10] C. R. Harvey, Y. Liu, and H. Zhu, "...and the cross-section of expected returns," *Rev. Financ. Stud.*, vol. 29, no. 1, pp. 5–68, 2016.
- [11] D. H. Bailey and M. López de Prado, "The deflated Sharpe ratio," *J. Portfolio Manage.*, vol. 40, no. 5, pp. 94–107, 2014.
- [12] D. H. Bailey, J. Borwein, M. López de Prado, and Q. J. Zhu, "The probability of backtest overfitting," *J. Comput. Finance*, vol. 20, no. 4, pp. 39–70, 2017.
- [13] H. White, "A reality check for data snooping," *Econometrica*, vol. 68, no. 5, pp. 1097–1126, 2000.
- [14] P. R. Hansen, "A test for superior predictive ability," *J. Bus. Econ. Stat.*, vol. 23, no. 4, pp. 365–380, 2005.
- [15] Y. Benjamini and Y. Hochberg, "Controlling the false discovery rate," *J. Roy. Stat. Soc. B*, vol. 57, no. 1, pp. 289–300, 1995.
- [16] T. Fischer and C. Krauss, "Deep learning with long short-term memory networks for financial market predictions," *Eur. J. Oper. Res.*, vol. 270, no. 2, pp. 654–669, 2018.
- [17] C. Krauss, X. A. Do, and N. Huck, "Deep neural networks, gradient-boosted trees, random forests: Statistical arbitrage on the S&P 500," *Eur. J. Oper. Res.*, vol. 259, no. 2, pp. 689–702, 2017.
- [18] S. Gu, B. Kelly, and D. Xiu, "Empirical asset pricing via machine learning," *Rev. Financ. Stud.*, vol. 33, no. 5, pp. 2223–2273, 2020.
- [19] Z. Zhang, S. Zohren, and S. Roberts, "DeepLOB: Deep convolutional neural networks for limit order books," *IEEE Trans. Signal Process.*, vol. 67, no. 11, pp. 3001–3012, 2019.
- [20] P. N. Kolm, J. Turiel, and N. Westray, "Deep order flow imbalance," *Math. Finance*, vol. 33, no. 4, pp. 1044–1081, 2023.
- [21] M. López de Prado, *Advances in Financial Machine Learning*. Hoboken, NJ, USA: Wiley, 2018.
- [22] C. J. Clopper and E. S. Pearson, "The use of confidence or fiducial limits illustrated in the case of the binomial," *Biometrika*, vol. 26, no. 4, pp. 404–413, 1934.
- [23] M. H. Pesaran and A. Timmermann, "A simple nonparametric test of predictive performance," *J. Bus. Econ. Stat.*, vol. 10, no. 4, pp. 1992.
- [24] G. Ke et al., "LightGBM: A highly efficient gradient boosting decision tree," in *Proc. NeurIPS*, vol. 30, 2017, pp. 3146–3154.
- [25] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. ACM SIGKDD*, 2016, pp. 785–794.
- [26] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Comput.*, vol. 9, no. 8, pp. 1735–1780, 1997.
- [27] A. Vaswani et al., "Attention is all you need," in *Proc. NeurIPS*, vol. 30, 2017, pp. 5998–6008.
- [28] B. Lim, S. Ö. Arik, N. Loeff, and T. Pfister, "Temporal fusion transformers for interpretable multi-horizon time series forecasting," *Int. J. Forecast.*, vol. 37, no. 4, pp. 1748–1764, 2021.
- [29] D. N. Politis and J. P. Romano, "A circular block-resampling procedure for stationary data," in *Exploring the Limits of Bootstrap*. New York, NY, USA: Wiley, 1992, pp. 263–270.
- [30] R. F. Engle, "Autoregressive conditional heteroscedasticity," *Econometrica*, vol. 50, no. 4, pp. 987–1007, 1982.
- [31] T. Bollerslev, "Generalized autoregressive conditional heteroskedasticity," *J. Econometrics*, vol. 31, no. 3, pp. 307–327, 1986.
- [32] P. Carr and L. Wu, "Variance risk premiums," *Rev. Financ. Stud.*, vol. 22, no. 3, pp. 1311–1341, 2009.
- [33] G. Bakshi and N. Kapadia, "Delta-hedged gains and the negative market volatility risk premium," *Rev. Financ. Stud.*, vol. 16, no. 2, pp. 527–566, 2003.